

What do these reports say about Java?

The insights drawn from the reports discussed above can be summarised in the following points:

1. Java remains a popular programming language in the developer community, although its popularity has seen a slight decline due to the rise of programming languages like Python and Rust.
2. Although Java is used in many industries, its ecosystem primarily consists of backend Web developers. This is in huge part due to the popularity and demand of the Spring Web development framework, which is the most popular Java based Web development framework.
3. The developer community is losing interest in learning Java or investing time into learning frameworks that require Java. The lack of interest in learning Hadoop and Spring is a case in point.
4. Most developers coding professionally find Java an intimidating language. This resistance spills over in the statistics for Java based frameworks as well. A significant majority of Hadoop developers dread using Hadoop, whereas a significant minority of Spring developers dread using Spring.

Summing it up

By no means is Java a dead programming language. But it is most certainly a programming language that is facing trouble finding its footing in the ever evolving developer community. In the beginning of the article, I had shared the design principles of Java as described by Oracle. As far as the utility of Java is concerned, Oracle has done a great job. Java can be used for a wide variety of applications efficiently. However, the price of this utility has been the sacrifice of simplicity. The first design

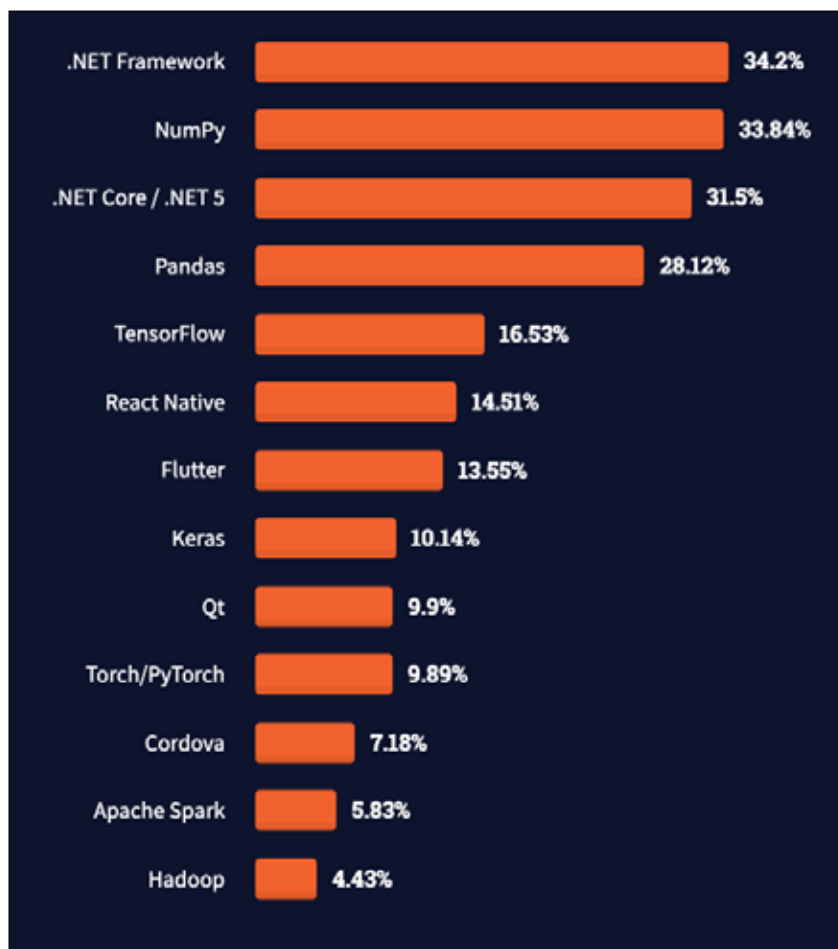


Figure 5: Key insights on Hadoop in the Stack Overflow Developer Survey 2021

principle of Java is that it should be *simple*. Unfortunately, developers do not seem to find it simple to use.

Most of the developers who are participating in these surveys have been working for 10 years or less. This new generation of developers is having trouble with the strict implementation of the object-oriented design pattern of Java. With the rising popularity of Python (thanks to its many libraries) and the immense rise of computational power that allows

many companies to not sweat over the time complexity of programs, Java is beginning to be seen more as an obstacle rather than a solution.

The beauty of Python and other such programming languages that do not strictly implement object-oriented programming is that due to their syntactic simplicity, it is easier to write semantically complex code with them.

So to conclude, I would say that Java is most certainly not dead, YET. **END** 🐧

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